

Boundless Magic

Game Design Document

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1. Introduction

Basic Description and Target Audience

Boundless Magic is a 2D side-scrolling endless runner for mobile, where the player evades obstacles, fights enemies and tries to escape from the dungeon and get the highest score. The player's character can cast multiple powerful spells, that aid him in various ways during the run. Spells behave like combos. To cast a spell, the player needs to kill certain enemy types on their way e.g. two red enemies and one blue, but killing enemies in sequence is not required. The player can choose and equip up to three of their favourite spells. The combinations for each spell are randomly picked from a predesigned set. During each run, the player is required to kill different set of enemies, but spell behaviour stays the same.

Boundless Magic is targeted for every endless runner lover and enthusiast of the genre's classics given below. We aim to captivate them by keeping well established gameplay and spicing it up with semi-tactical combat.

Inspiration

We were inspired by mobile endless runners such as *Subway Surfers*, *Jetpack Joyride* and *Temple Run*, which we consider as classics.

2. Gameplay

Goal and Lose Conditions

The goal of the game is to achieve the highest score. Player learns to be better, unlocks new spells and upgrades them to eventually beat his own highest score.

To fail, the player must:

- Directly run into the wall
- Directly run into the enemy
- Hit an enemy or a wall during *the last chance* (see: Movement) period

If one of these conditions is met, the run ends.

Movement

The player's character is constantly running to the side of the screen. In the playable area, there are four floors. The player moves between them by going upwards and downwards. While being on the highest floor, the player can move upwards and traverse to the lowest floor and vice versa.

As player progresses through the run, character's running speed increases. The run begins with fixed speed and only after defeating few enemies, the speed begins to very slowly increase to a specified cap¹. The player's character runs with this speed until another condition is met. These conditions might be as simple as the number of defeated enemies or an exact score. Only after meeting another condition, the player's character speed further rises. The speed can increase just to a certain point and when this limit is reached, the speed no longer increases. Maximum speed will be definitely challenging for the player. These measures are taken to: 1) make progressing during the run little longer, so the player won't reach the max speed too quickly; 2) skill check the player.

The player's character hitboxes are not entirely merciless. A playable character has two hitbox zones: forgiving and unforgiving. Forgiving zone is located on the edges of a character (tips of hands, legs, head etc.) Unforgiving zone is located more to the character's core. This breakdown makes the gameplay less frustrating, because when the player's character just brushes against a wall or an enemy, the game won't instantly end. Brushing against an obstacle triggers *the last chance*. During this period any other direct or indirect hit will result in player's failure. This period lasts for 8 seconds.

Walls

A wall (along with an enemy) is a type of an obstacle that player encounters during a run. The player needs to evade the walls, because shooting does not damage them.

¹ Details of running speeds, duration of each speed range, speed caps and conditions mentioned in this paragraph need to be tested.

Fighting Enemies

During the run, enemies spawn along with the walls in pre-designed sets, also called chunks (see: Procedurally Generated Obstacle Sets). A player can shoot projectiles, that kill the enemies. All enemies can be killed instantly with only one projectile.

There are 3 types of enemies: red, blue and yellow. After killing an enemy, the player receives an element of its' colour. Elements are used for casting spells (see: Spells).

Spells (combos)

During the run, in addition to shooting projectiles, the player can also cast powerful spells, that aid him and let him get higher score.

When the player enters a game for the first time and intends to begin a new run, the game shows him a screen with his equipped spells. In general, there are 3 slots and only one spell can be equipped in each slot. But to introduce the player into the game, at first, there is only one slot, that is already equipped with a starting spell².

The player needs to press a "Draw" button. The game randomizes the elements combination (combo sequence) for the equipped spell. There are 10 elements combinations, that the randomize system can choose from:

Combination 1	Blue	Red	Yellow
Combination 2	Blue	Blue	Blue
Combination 3	Red	Blue	Blue
Combination 4	Red	Red	Red

² For more information about upgrading slots and unlocking new spells, see Progression and Meta-Game.

Combination 5	Yellow	Red	Yellow
Combination 6	Yellow	Yellow	Yellow
Combination 7	Yellow	Blue	Blue
Combination 8	Red	Red	Yellow
Combination 9	Red	Red	Blue
Combination 10	Blue	Yellow	Yellow

After the system assigned a combination for the spell, the run begins.

Next, by counting which colours appear most frequently and which appear least frequently in the combinations, the system determines the probability of each enemy's colour appearing at spawn points. For example, the system randomized one combination for one equipped spell; the combination is: 1x yellow, 2x blue. This means, the chance of spawning an enemy of particular colour will be:

- 50% for blue enemy;
- 30% for yellow enemy;
- 20% for red enemy.

How the system calculates the spawn rates needs further designing, testing, so TBD.

In the Fighting Enemies section, we mentioned that after killing an enemy, the player receives an element of killed enemy colour. For example: 1) the player shoots a projectile; 2) the projectile hits a blue enemy; 3) the player receives a blue element. Gathered elements are clearly visible to the player³.

³ Current approach is to display elements just beside player's character head, but it needs further testing.

To cast a spell, the player needs to collect all the elements of a combination that is assigned to this spell. Continuing our example, the player has to gather 1x yellow and 2x blue elements to cast a spell. This means that a player needs to kill one yellow enemy and two blue enemies. **Important: the order of gathering colours doesn't matter.** When the player gathers all required elements, the spell is ready to be cast.

If the player gathers a combination that is not assigned to any of the equipped spells, the collected elements delete instantly after receiving the last third element.

The player can hold only one ready to be cast spell⁴. But while having a ready to be cast spell, the player can still gather elements.

List of spells:

- **Invulnerability** – the player's character changes its' appearance slightly and becomes invulnerable for 5 seconds. Running into a wall or an enemy does nothing and the player can't fail the run, while the effect lasts. There is also a warning duration, which is signaled by player's character blinking (invulnerable and normal appearance alternately). It triggers when only 2 seconds remains until the end of the effect.
- **Fireball** – the player's character shoots a powerful projectile that flies to the right and destroys all walls and enemies on its' way. Duration and details TBD.
- **Time slow** – for a short time, the player's character instantly runs slower, giving the player more time to react. When the effect is about to wear off, the running speed gradually increases to its' previous value. Details TBD.
- **Shield** – the player's character gets a shield that destroys walls, but disappears when hitting an enemy. Details TBD.
- **Curse** – immediately after casting, kills all enemies on screen Details TBD.
- **Sword** – the player's character receives a sword that leans to his right. It can kill all enemies instantly when hit but breaks if it touches the wall. Details TBD.

⁴ Right now, there isn't any solution on how to display a ready to be cast spell in the UI. So, it's now TBD.

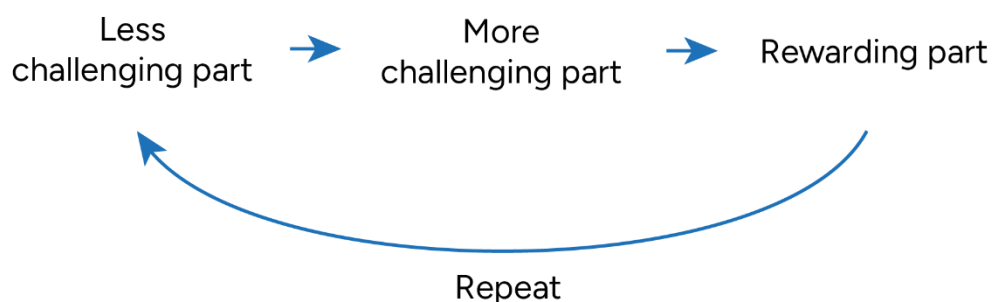
- **Triple-trouble** – for some amount of time, two shadows of the player's character appear on adjacent floors (one shadow on each floor) and earn points for the player. Shadows are not hurt by an enemy or a wall. Details TBD.

Procedurally Generated Obstacle Sets

To be more in control of what challenge the player faces, Boundless Magic uses procedural generated obstacle sets. A set can consist of: walls, enemy spawn points and collectible items. The layouts of these sets are predesigned, but the system that determines which set to spawn next is procedural.

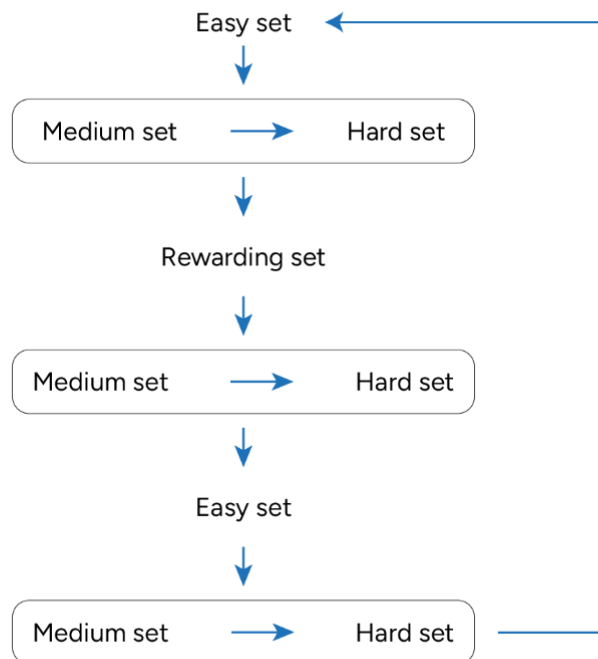
Spawn point for an enemy is an entity, that was placed into a set during the design of a set. When a set is spawned into the game world, the system, based on enemy spawn rates, spawns enemies. Enemy spawn rates example is mentioned in the Spells section of this document. They were: 50% for blue, 30% for yellow, 20% for red spawn chances. So, following this example, there is 50% chance that a blue enemy spawns on the place of enemy spawn point in each set that is placed into the game world.

The basic concept on which the set spawning system operates is:



This loop can be considered as an element of core gameplay loop. Firstly, the player faces a less challenging part to gradually introduce themselves into the run. Secondly, more challenging part is thrown at the player to slightly raise the stakes. Lastly, the player confronts rewarding part to keep him engaged. Rewarding part and less challenging part gives the player some rest and let him catch a breath in some way.

This solution can be further broken down into smaller parts. We plan to divide sets into four groups: 1) easy; 2) medium; 3) hard; and 4) rewarding. This division is made to have even more control of challenges presented to the player. The loop might also look like this:



To make the gameplay less repetitive and give it an illusion of randomness slight alterations and deviations of this loop might be possible, but details need to be designed and tested. This also applies to the rewarding set, because gratification needs to be very well balanced. TBD.

Every encounter with the enemies within the set is designed in a way that gives the player a choice whether to kill an enemy or evade it. We want to make sure that the player feels in control of picking which enemy to kill and collect an element, that they actually want.

Score

Just running and staying alive rewards the player with a small number of points⁵. Killing an enemy gives the player 10 points.

⁵ Details on how many points the player will earn needs to be tested, so it's TBD.

But to encourage the player to cast spells and not avoid enemies, the highest number of points can be earned by casting spells. TBD: A table describing how many points each spell gives, will be listed here.

Progression and Meta-Game

During the run, the player can collect special items, that will be used for upgrading his spells and buying more slots for equipping spells. More details on these items TBD.

There are 3 slots that can be equipped with a spell, as mentioned in the Spells section. At first only one slot is available, though the player can quickly unlock another slot to keep him engaged. This solution ensures that the player understands core gameplay mechanics of spell casting, before giving him more options. Price for the upgrade and details TBD.

The player can also unlock new spells and upgrade those already owned. But all the details need to be designed, so TBD.

Controls

Boundless Magic is targeted to be played on mobile in a horizontal view.

To move the character up and down, the player needs to briefly swipe with their finger upwards or downwards.

To shoot a projectile, the player needs to tap the screen.

To cast a spell, the player needs to quickly tap the screen twice

or

tap a dedicated button in a corner of the screen (it needs to be tested).